



# Hybrid classical-quantum multilayer neural networks for monitoring agricultural activities using remote sensing data

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Received: 28 September 2024 / Accepted: 19 December 2024 / Published online: 4 January 2025  
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## Abstract

In recent years, machine learning (ML), supported by the surge in computational power, has been integrated into numerous research domains, providing vital solutions to a broad spectrum of challenges. However, the increasing volume and complexity of data utilized by ML systems have necessitated the exploration of new methods to quicken and refine the precision of existing ML algorithms. The utilization of satellite data in agriculture has grown extensively, providing numerous opportunities to improve different facets of agricultural management and productivity. Interestingly, the volume of data produced increases exponentially with time posing a challenge for classical ML computing methods. Quantum computers, capable of exploiting an exponentially vast quantum state space via quantum superposition and entanglement, are aptly suited to handle such data volumes. Therefore, the integration of classical ML algorithms with quantum computing, known as quantum machine learning (QML), holds significant promise in processing satellite data. In our research, we developed, trained, and evaluated novel hybrid classical-quantum multilayer neural networks, comparing their efficacy with conventional classical networks. We applied QML to two classification tasks, namely the classification of rice and cotton crops using satellite data obtained by Sentinel-1 and Sentinel-2 satellites. Our results show that integrating quantum neurons into classical multilayer neural networks substantially improves their performance.

**Keywords** Quantum computing · Quantum machine learning · Quantum neural networks · Remote sensing · Classification · Satellite imagery · Satellite data

## 1 Introduction

As we are going through the third decade of the twenty-first century, machine learning (ML) still plays a crucial role alongside other artificial intelligence (AI) technologies, such as natural language processing (NLP) (Nagarhalli

et al. 2021; Nadkarni et al. 2011) and computer vision (Sebe 2005; Khan and Al-Habsi 2020; Khan et al. 2021). Machine learning models are widely utilized to classify, cluster, and recognize patterns inside large volumes of data. ML models are used in a wide spectrum of applications, ranging from agriculture, economics, and healthcare, to name only a few. There is a trend among the largest global 2000 corporations (Gomez-Urbe and Hunt 2015; Parveen and Varma 2021; Mahesh 2020) for using ML-driven solutions as part of their product development and service providing systems.

Quantum computers are a significant asset in tackling modern, difficult-to-solve problems. Grover's (Grover 1996) and Shor's (Shor 1999) algorithms are the two most known algorithms that demonstrate the potential benefit quantum computers could have over their classical counterparts. By exploiting quantum phenomena such as the quantum superposition, interference, and entanglement, they can efficiently find solutions to problems believed to be too complicated to be solved by novel classical computers (Dam et al. 2006;

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Hallgren 2007). The integration of quantum computers with classical ML algorithms, also known as quantum machine learning (QML), can leverage those quantum effects and lead to quantum speed-ups over classical ML algorithms (Schuld and Petruccione 2018; Biamonte et al. 2017; Mitarai et al. 2018). Until recently, a number of QML models have been developed such as quantum support vector machines (QSVMs) that utilize quantum kernels (Jäger and Krems 2023), variational circuit quantum classifiers (Havlíček et al. 2019), and quantum neural networks (Farhi and Neven 2018; Arthur 2022).

The increasing demand for food supply is a major problem that humanity faces, implied by the growth of the world population. Remote sensing techniques are increasingly utilized in agriculture and agronomy, providing a distinct viewpoint on the spatial and temporal dynamics of agricultural systems. Monitoring agricultural activities presents unique challenges, including pronounced seasonal patterns tied to the biological life cycle of crops, reliance on physical landscape and climate variables, and significant variability across both space and time. One of the proposed solutions is the optimization of agricultural crops to produce a greater supply of food, while minimizing the environmental impact. In this context, many scientists rely on remote sensing data, to create valuable ML models to assist them in the decision-making process upon the agricultural crops' optimization (Kussul et al. 2017; Orynbaikyzy et al. 2019). The study of ML has focused on resolving this global, societal problem and consequently QML (Nguyen and Wagner 2017; Bauer-Marschallinger et al. 2018) can now be incorporated.

Within our research, we developed, trained, and evaluated quantum circuits as part of multilayer feedforward neural networks (MLFFNNs) and applied to classification tasks. For this purpose, we created, trained, tested, and evaluated a number of hybrid classical-quantum MLFFNNs (HCQNNs) model variations and compared their performance to classical MLFFNNs and other well-known classical classification

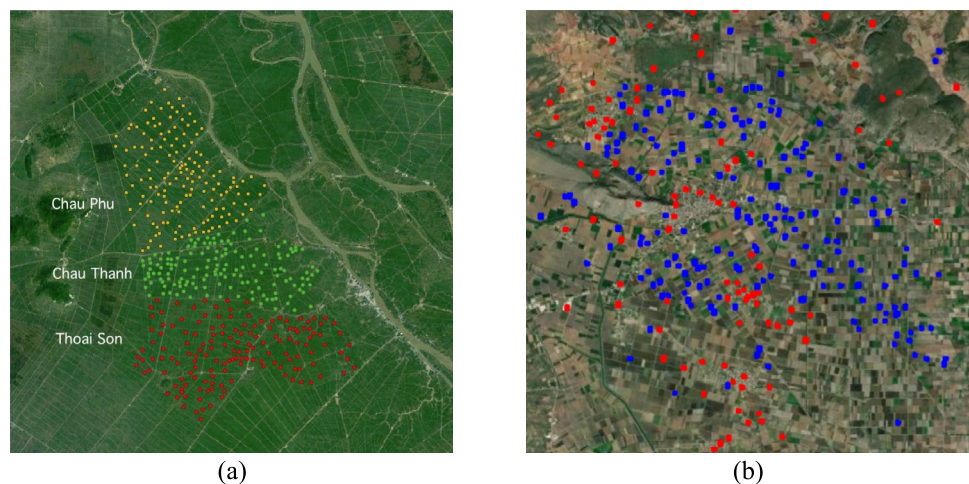
models, such as the Scikit-learn's support vector classifier and gradient boosting classifier (Pedregosa et al. 2011). For all model variations and comparisons of the aforementioned models, we utilized two remote sensing binary classification datasets, one for rice crops and one for cotton crops. We compared the performance of our models to the performance of well-known classical models and to the performance of quantum support vector machine classifier (QSVC). Our results show that integrating quantum neurons into MLFFNNs significantly enhances their performance using a small number of qubits. In most cases, our HCQNNs outperform both the classical models and the QSVC as shown in Tables 2, 3, 4, 5, and 6.

## 2 Satellite datasets

As stated previously, to evaluate the performance of our hybrid classical-quantum ML models and compare their results with classical ML models, we created two remote sensing classification datasets, namely the "Rice Crop" and "Cotton Crop" datasets. These datasets comprise data originating from two of the Copernicus project satellites, namely the Sentinel-1 and Sentinel-2 satellites (Copernicus programme: The Sentinel-1 mission n.d. ; Copernicus programme: The Sentinel-2 mission n.d. ). The collected data are for the years 2021–2022 for the "Rice Crop" and 2021–2023 for the "Cotton Crop" datasets. The data is publicly available via Microsoft's Planetary Computer portal (Planetary and Computer: A Planetary Computer for a Sustainable Futurehttps:, , planetarycomputer.microsoft.com, . n.d.).

The "Rice Crop" is a five-feature dataset used to classify parcels into Rice (1) and Non-Rice (0) areas in the An Giang province in the Mekong Delta, Vietnam shown in Fig. 1a. The five features comprising this dataset are as follows: (1) vertical-vertical (VV), (2) vertical-horizontal (VH)

**Fig. 1** **a** The An Giang province in the Mekong Delta in Vietnam. This area consists of three central districts (Chau Phu, Chau Thanh, and Thoi Son) (EY 2023). **b** The wider area of Orchomenos, Levadia, Greece. Areas denoted in blue color rectangles comprise cotton crop parcels, whereas the ones in red color rectangles comprise non-cotton crop areas



polarization band data, (3) normalized difference vegetation index (NDVI), (4) difference vegetation index (DVI), and (5) modified soil ratio (MSR). The VV and VH express the value of polarization between the transmitted and received radar signal on the Sentinel-1 satellite, whereas the NDVI, DVI, and MSR are commonly known vegetation indices, often used in remote sensing applications (Bannari et al. 1995; Huete 2012). These vegetation indices are statistical combinations of the “filtered” spectral bands data derived from Sentinel-2. The “Rice Crop” dataset comprises 750 samples, 600 of which were utilized in a 0.8/0.2 train/validation split for the training and the rest were utilized for testing purposes.

The “Cotton Crop” is a seven-feature dataset used to classify parcels as cotton (1) and non-cotton (0) areas in the wider area of Orchomenos, Levadia, Greece, shown in Fig. 1b. To create this dataset, we utilized publicly available data derived from the Agriculture Cooperative of Orchomenos (Agriculture Cooperative of Orchomenos n.d). The seven features comprising this dataset are as follows: (1) vertical-vertical (VV), (2) vertical-horizontal (VH) polarization band data, (3) Y measuring luminance of the image, (4) Cb measuring the chrominance value that indicates the difference between the blue component and a reference value, (5) Cr measuring the chrominance value that indicates the difference between the red component and a reference value, (6) normalized difference green/red difference index (NGRDI) (Hamuda et al. 2016), and (7) modified green/red vegetation index (MGRVI) (Bendig et al. 2015). The NGRDI and MGRVI are well-known vegetation indices, often used for cotton crop parcel identification, whereas the Y, Cb, and Cr are color space data. As in the vegetation indices, the color space data mentioned are also statistical combinations of the “filtered” spectral bands data derived

from Sentinel-2. The “Cotton Crop” dataset comprises 450 samples, 300 of which were utilized for training and the rest were utilized for testing purposes. For this case, we did not include validation during the models’ training phase since this dataset’s size is not large.

### 3 Structure of the MLFFNN architectures

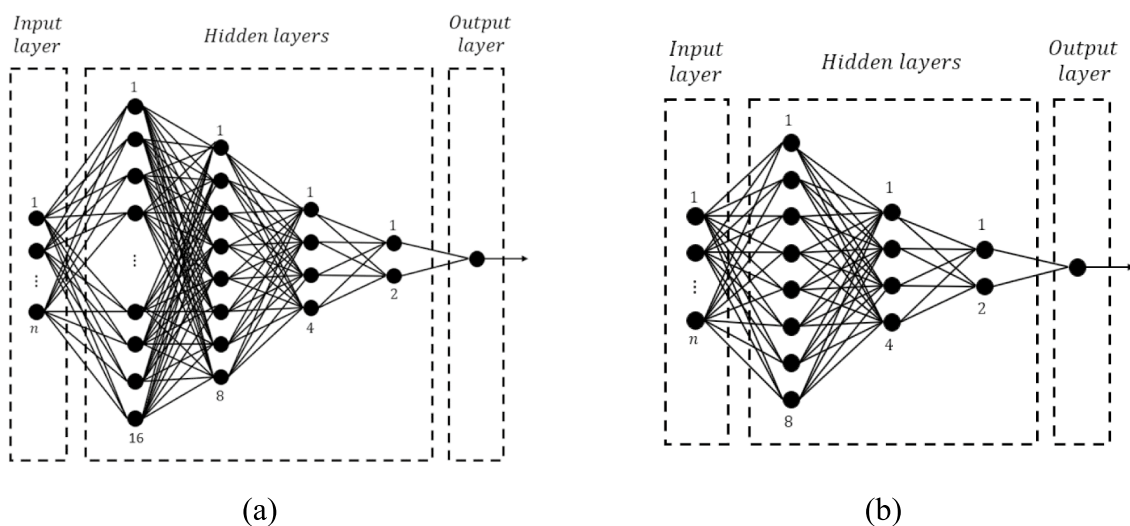
We developed a large number of neural network variations to classify the data in the two datasets, using both classical and hybrid classical-quantum approaches. We focused on multi-layer feedforward neural networks (MLFFNNs) for both our model’s types. To compare the classical and hybrid classical-quantum models, we adopted the same model structures for both cases. The two MLFFNN classical architectures used throughout this work are shown in Fig. 2.

The Model 1 architecture (Fig. 2a) comprises four hidden layers, each one consisting of 16, 8, 4 and 2 neurons from left to right, whereas the Model 2 architecture (Fig. 2b), comprises three hidden layers, consisting of 8, 4, and 2 neurons from left to right. Henceforth, these architectures will be referred to as Model 1 and Model 2 accordingly.

#### 3.1 Hybrid classical-quantum neural network (HCQNN) structures

As previously mentioned, we created and tested 83 HCQNN structures, shown in the Supplementary Information. To be easily distinguishable, these structures can be divided into three subcategories:

1. HCQNNs with quantum layers



**Fig. 2** **a** Model 1 architecture comprising four hidden layers. **b** Model 2 architecture comprising three hidden layers

2. HCQNNs with 1 quantum neuron as part of a single hidden layer
3. HCQNNs with 2 quantum neurons as part of:
  - a. Different hidden layers
  - b. The same hidden layer

The term “Quantum Neuron(s)” refers to quantum circuit(s) acting as parts of the classical neural network. The aforementioned structures were developed using PyTorch (Paszke et al. 2019) and Qiskit (Javadi-Abhari et al. 2024). The interconnection between the classical and the quantum part of the neural network was achieved using the PyTorch Torchconnector class. Each quantum circuit comprises the Feature Mapping circuit and the Ansatz. The Feature Mapping circuit encodes the classical data into quantum states, and the Ansatz is a parameterizable and trainable circuit, which classifies the data. We tested eight different combinations of Feature Mapping circuits and Ansätze, shown in Table 1. In most cases, the “ZZFeatureMap” (Havlíček et al. 2019; Javadi-Abhari et al. 2024) and the “RealAmplitudes”

(Javadi-Abhari et al. 2024) quantum circuits were utilized as the Feature Mapping and Ansatz circuit, respectively, since they are some of the most well-known and widely used quantum circuits for classification purposes. The “ZZFeatureMap” and the “RealAmplitudes” quantum circuits are shown in Fig. 3a and b, whereas the overall quantum circuit is presented in Fig. 3c.

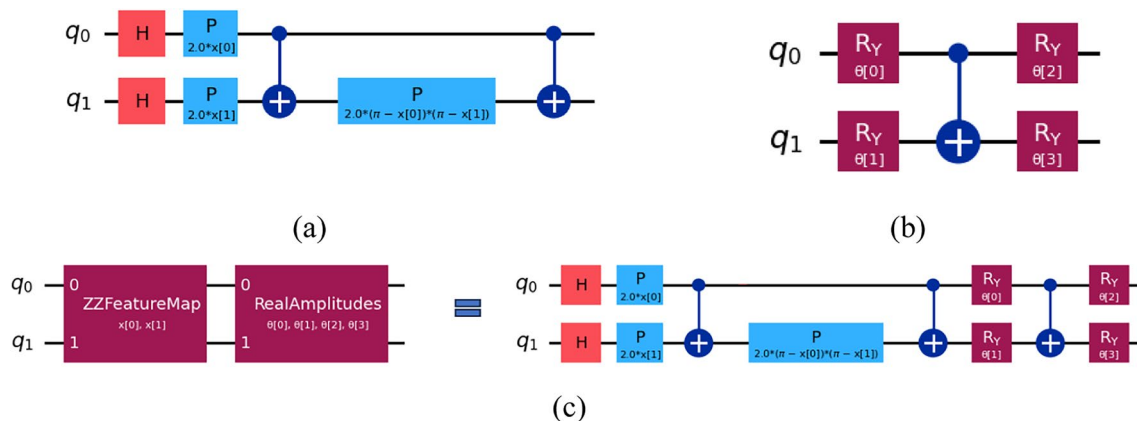
The entanglement structures we selected for the “ZZFeatureMap” and “RealAmplitudes” quantum circuits were the “full” and “reverse linear,” respectively. More specifically, the “full” entanglement structure corresponds to the state where each qubit is entangled with all the other qubits of the quantum circuit. On the other hand, the “reverse linear” entanglement structure refers to the state where each qubit  $i$  is entangled with qubit  $i + 1$ , for all  $i \in \{n - 2, n - 3, \dots, 1, 0\}$  where  $n$  is the total number of qubits in our quantum circuit. To further clarify the aforementioned entanglement structures, we demonstrate in Fig. 4a and b the “ZZFeatureMap” and “RealAmplitudes” quantum circuits, each one consisting of four qubits.

In addition to the aforementioned circuits, we tested quantum circuits 14 and 15 that offer entangling capability of parameterized quantum circuits for hybrid classical-quantum algorithms (Sim et al. 2019), both as Feature Mapping circuits and as Ansätze, as shown in Table 1. Thus, we explored how the expressibility of variational quantum circuits, i.e., their ability to generate (pure) states that are well representative of the Hilbert space (Sim et al. 2019), affects the HCQNN’s overall performance. The quantum circuits 14 and 15 are shown in Fig. 5a and b, respectively.

We also developed a much simpler method to introduce the features to the quantum circuit, by removing the entanglement existed in the aforementioned circuits and encoding our features as a linear combination of probability amplitudes. We directly opposed this method with circuit 14 in terms of how the expressibility may affect

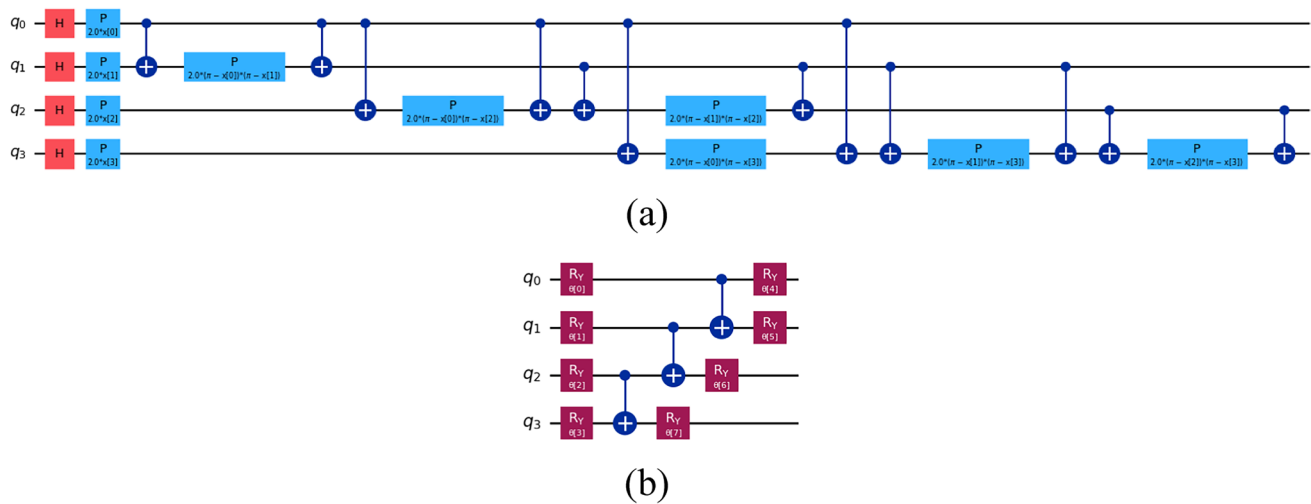
**Table 1** All Feature Map and Ansatz circuit combinations alongside their structure subcategory in which they were tested

| Feature Map    | Ansatz         | Subcategory |   |    |    |
|----------------|----------------|-------------|---|----|----|
|                |                | 1           | 2 | 3a | 3b |
| ZZFeatureMap   | RealAmplitudes | ✓           | ✓ | ✓  | ✓  |
|                | Circuit 14     | ✗           | ✓ | ✗  | ✗  |
|                | Circuit 15     | ✗           | ✓ | ✓  | ✗  |
| AngleEmbedding | RealAmplitudes | ✗           | ✓ | ✓  | ✓  |
|                | Circuit 14     | ✗           | ✓ | ✗  | ✓  |
|                | Circuit 15     | ✗           | ✓ | ✓  | ✗  |
| Circuit 14     | RealAmplitudes | ✗           | ✓ | ✗  | ✗  |
| Circuit 15     | RealAmplitudes | ✗           | ✓ | ✗  | ✗  |



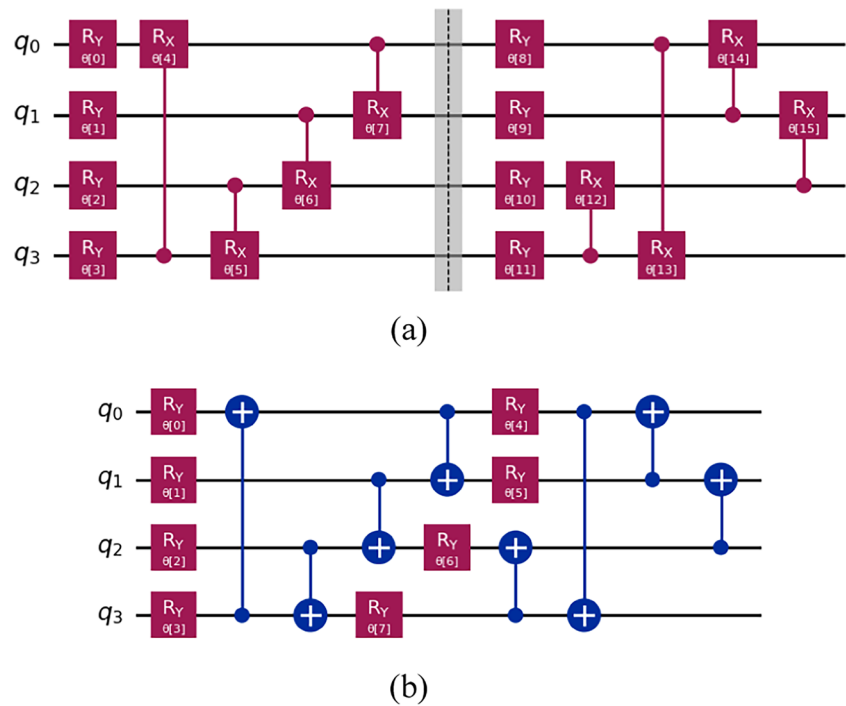
**Fig. 3** a The “ZZFeatureMap” quantum circuit. b The “RealAmplitudes” quantum circuit. Both these circuits comprise 2 qubits, with each circuit to be repeated once. c The overall quantum circuit comprising the “ZZFeatureMap” and the “RealAmplitudes” quantum circuits





**Fig. 4** **a** The “ZZFeatureMap” quantum circuit, with “full” entanglement structure. **b** The “RealAmplitudes” quantum circuit, with “reverse linear” entanglement structure. Both these circuits comprise four qubits, with each circuit to be repeated once

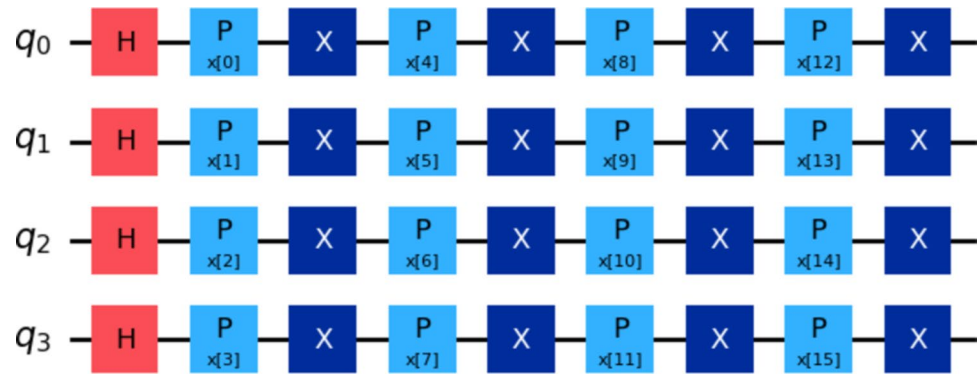
**Fig. 5** **a** Quantum circuit 14. **b** Quantum circuit 15



the model’s performance, given that for the same number of qubits, the same number of features are introduced. Additionally, we compared these circuits’ efficacy with “ZZFeatureMap” and circuit-15, which for equal number of qubits they introduce less features. We named this method “AngleEmbedding,” as shown in Table 1, since it relies strongly on the Angle Embedding technique (Lloyd et al. 2020). Our investigation showed that the “ZZFeatureMap” yielded better results than the

“AngleEmbedding,” and thus entanglement is essential, as described below. The number of qubits needed for the “AngleEmbedding” method is chosen as  $\log_2(L)$ ,  $L$  being the layer input size. Since each quantum circuit substitutes a classical neuron within the HCQNN, the number of features that needs to be encoded relies strongly on the previous layer output size, i.e., the number of classical neurons. Figure 6 shows the “AngleEmbedding” quantum circuit for layer input size 16.

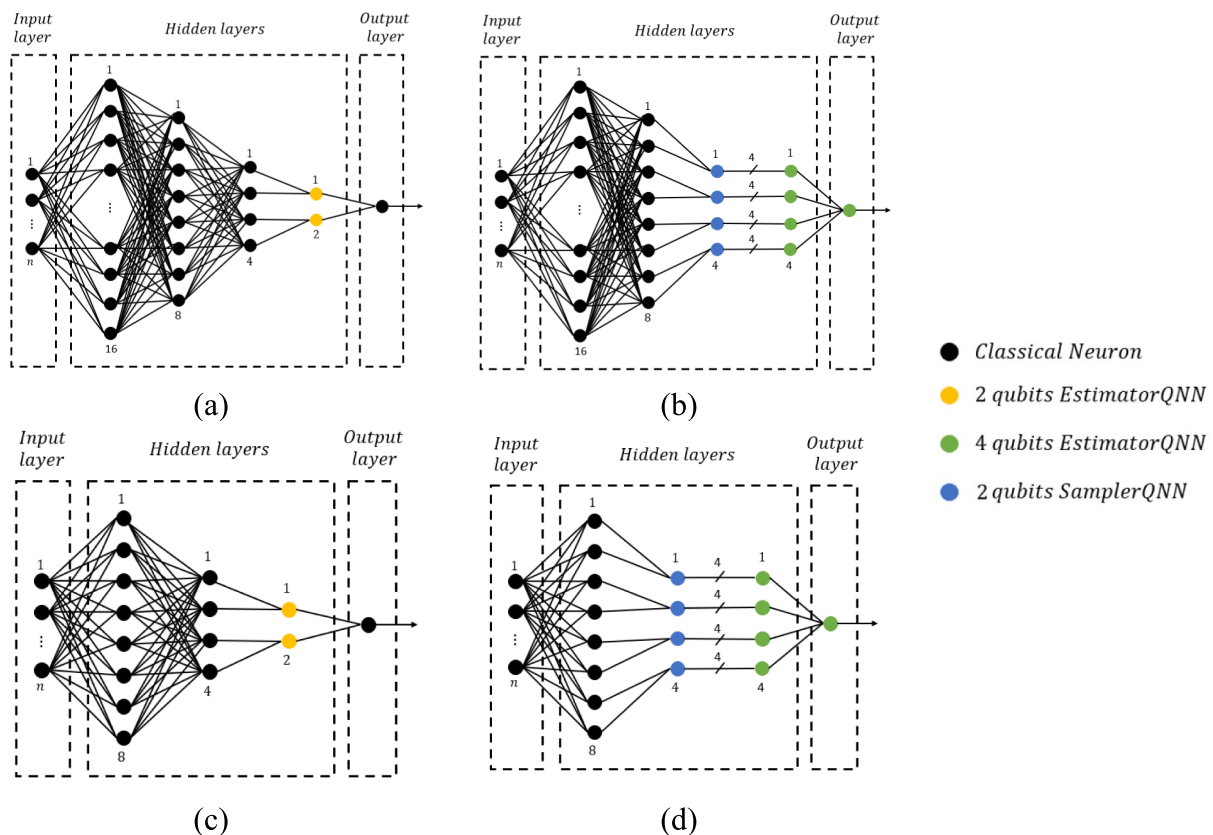
**Fig. 6** The “AngleEmbedding” circuit for layer input size 16. Only four qubits are required to properly introduce our data to the Ansatz



### 3.2 HCQNNs with quantum layers

To transform MLFFNN classical (hidden and output) layers into quantum ones we introduced quantum circuits into the aforementioned two classical models to create a hybrid classical-quantum structure, yielding four different HCQNN variations.

These quantum circuits comprise the Feature Mapping circuit and the Ansatz. For this case, we utilized the “ZZFeatureMap” as our Feature Mapping circuit and the “RealAmplitudes” circuit as our Ansatz. Moreover, to acquire the states of the qubits at each iteration, we utilized the IBM Qiskit’s Estimator and Sampler primitives, as shown in Fig. 7, which depicts the four different variations of the architecture. Both these primitives are low-level classes,



**Fig. 7** Four different HCQNN architecture variations that comprise estimator and sampler QNNs. **a** Four hidden layers with two qubit estimator QNNs that form one layer. **b** Four hidden layers with two qubit sampler QNNs and four qubit estimator QNNs that form two of the four layers and a four-qubit estimator QNN that forms the out-

put layer. **c** Three hidden layers with two qubit estimator QNNs that form one layer. **d** Three hidden layers with two qubit sampler QNNs and four qubit estimator QNNs that form two of the three layers and a four-qubit estimator QNN that forms the output layer

developed by IBM, allowing users to execute their quantum circuits both on simulators and on real IBM's quantum hardware. More specifically, the Sampler primitive provides the opportunity each user to execute quantum circuits and obtain their quasi-probability distribution, after multiple shots, in the  $Z$  measurement basis. Thus, the output shape of the SamplerQNN, i.e., the quantum circuit that makes use of the Sampler primitive, is equal to  $2^n$  where  $n$  is the number of the quantum circuit's qubits. On the other hand, the Estimator primitive allows the evaluation of a quantum circuit's expectation value when an observable is provided. Within the scope of this work, for the Estimator primitive the observable  $Z^{\otimes n}$  is selected, where  $n$  is the number of the corresponding quantum circuit's qubits.

### 3.3 HCQNNs with one quantum neuron as part of a single hidden layer

We also developed HCQNNs with one quantum neuron replacing a classical neuron in one layer, i.e., we used a quantum circuit instead of a classical neuron inside a NN's hidden layer. As mentioned before, this quantum circuit acts similarly to a classical neuron, and thus, we refer to this circuit as "quantum neuron." Furthermore, the number of qubits of each quantum circuit depends both on the previous layer's classical neurons (assuming that each neuron has a single output) and the encoding method. Like before, the observable  $Z^{\otimes n}$ , where  $n$  is the number of the corresponding quantum circuit's qubits, is considered.

By leveraging only a single quantum neuron in the entire NN, we were able to leverage fewer qubits, which has the positive benefit of producing quicker training and testing for our models. The HCQNN models consist of 4 and 3 hidden layers, and there are 14 different locations where the quantum neuron can replace the classical one. We created, tested, and evaluated all possible locations rising from the previous observation as depicted in Figures A1 and A4 in

the Supplementary Information. Figure 8 shows one of these variations.

### 3.4 HCQNNs with two quantum neurons as parts of hidden layers

We developed HCQNNs with two quantum neurons replacing two classical neurons in one or two hidden layers and evaluated the performance of these HCQNN architectures as shown in Figs. 9 and 10. With two quantum neurons located either in different layers or in the same layer, many different HCQNN variations arise. All these variations are shown in Figures A2, A3, A5, and A6 in the Supplementary Information. For all the cases of one and two quantum neurons, the IBM Qiskit's Estimator primitive was utilized to obtain the state of the qubits in each iteration.

## 4 Results and comparison

To train the aforementioned neural network variations, we utilized the adaptive moment estimation optimizer formally known as ADAM optimizer (Kingma and Ba 2014) with a learning rate equal to 0.008. We also selected the mean square error function (MSELoss) as the loss function (Paszke et al. 2019) used to train our models. Generally, for machine learning classification tasks, the cross-entropy loss is mostly favorable. However, this recommendation is given under the assumption that the output of the neural network ranges between 0 and 1. This is achieved by adding an activation function in the NN's output layer. In our case, we did not include any activation function to all the aforementioned HCQNNs output layers. Thus, our networks' output may not lie within the  $[0,1]$  range. This is the reason why the MSELoss function was selected to train our models, despite it not being the norm. The corresponding training accuracy and loss curves, for 20 training epochs, for the two

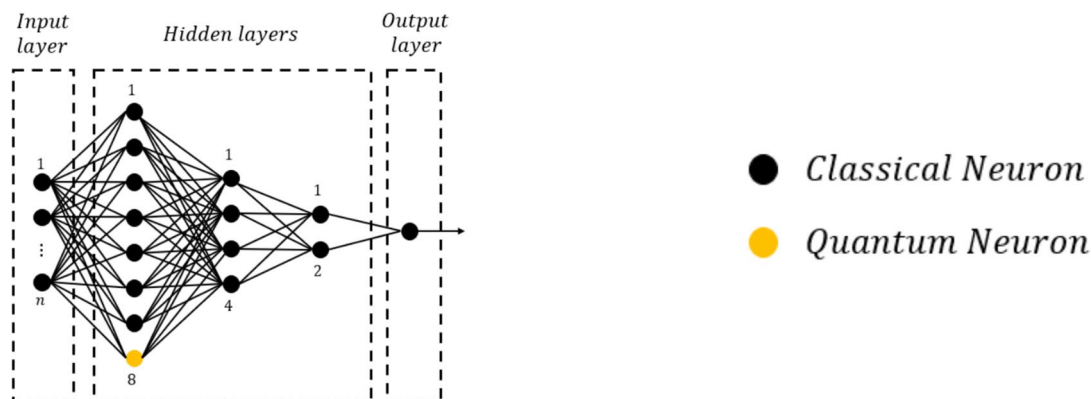
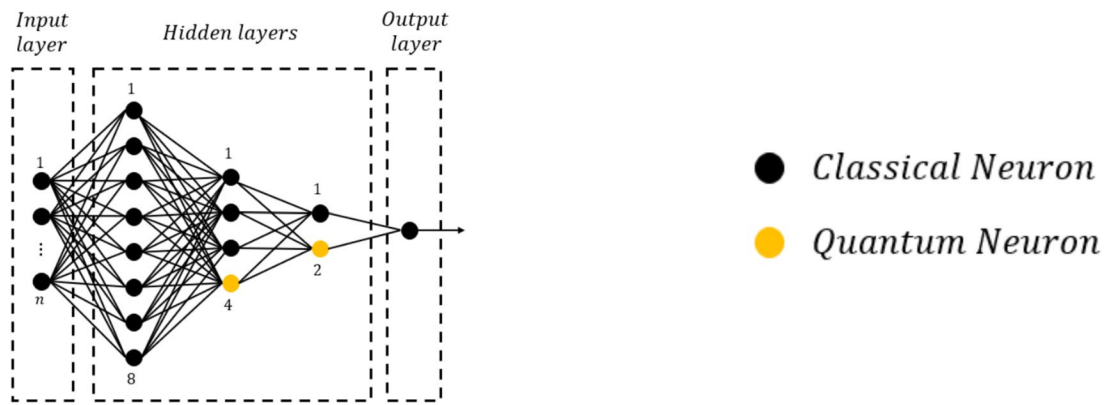
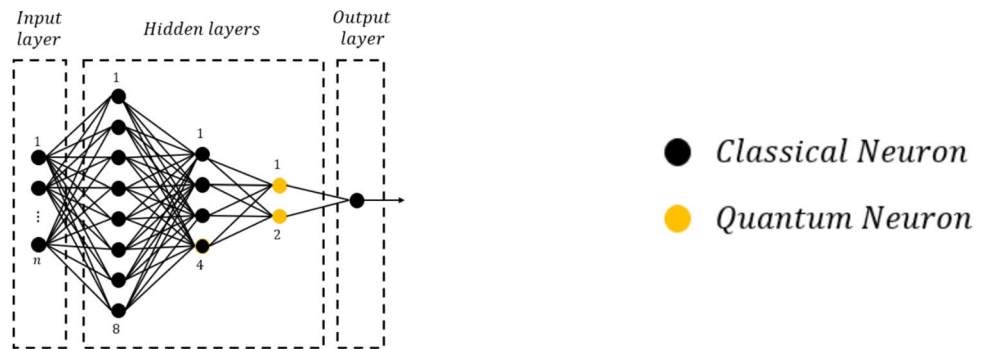


Fig. 8 HCQNN with one quantum neuron as part of a single hidden layer



**Fig. 9** HCQNN with two quantum neurons as part of two different hidden layers in the three hidden layer architectures

**Fig. 10** HCQNN with two quantum neurons as part of the same hidden layer in the three hidden layer architectures



best HCQNN models and the best classical MLFFNN model are presented in Fig. 11a and b. Finally, the F1 score metric was chosen to evaluate the performance of all the classical and hybrid classical-quantum neural network architectures as described in Eq. (1).

$$F1score = 2 * \frac{\frac{t_p}{(t_p+f_p)} * \frac{t_p}{(t_p+f_n)}}{\frac{t_p}{(t_p+f_p)} + \frac{t_p}{(t_p+f_n)}} \quad (1)$$

In Eq. (1),  $t_p$  is the number of the “true positive” samples, i.e., samples classified correctly to the 2nd class,  $f_p$  is the number of the “false positive” samples,  $t_n$  is the number of the “true negative” samples, i.e., samples classified correctly to the 1st class and  $f_n$  is the number of the “false negative” samples.

Next, we compare the performance of the hybrid classical-quantum models with the performance of well-known classical models. We applied 15 classical models to the satellite datasets described in Sect. 2. The performance (scores) of the classical models for both datasets is shown in Table 2. We also present information regarding those results’ variance in Fig. 12a and b.

We also applied classical MLFFNNs, which have the same architecture as our hybrid classical-quantum models, to

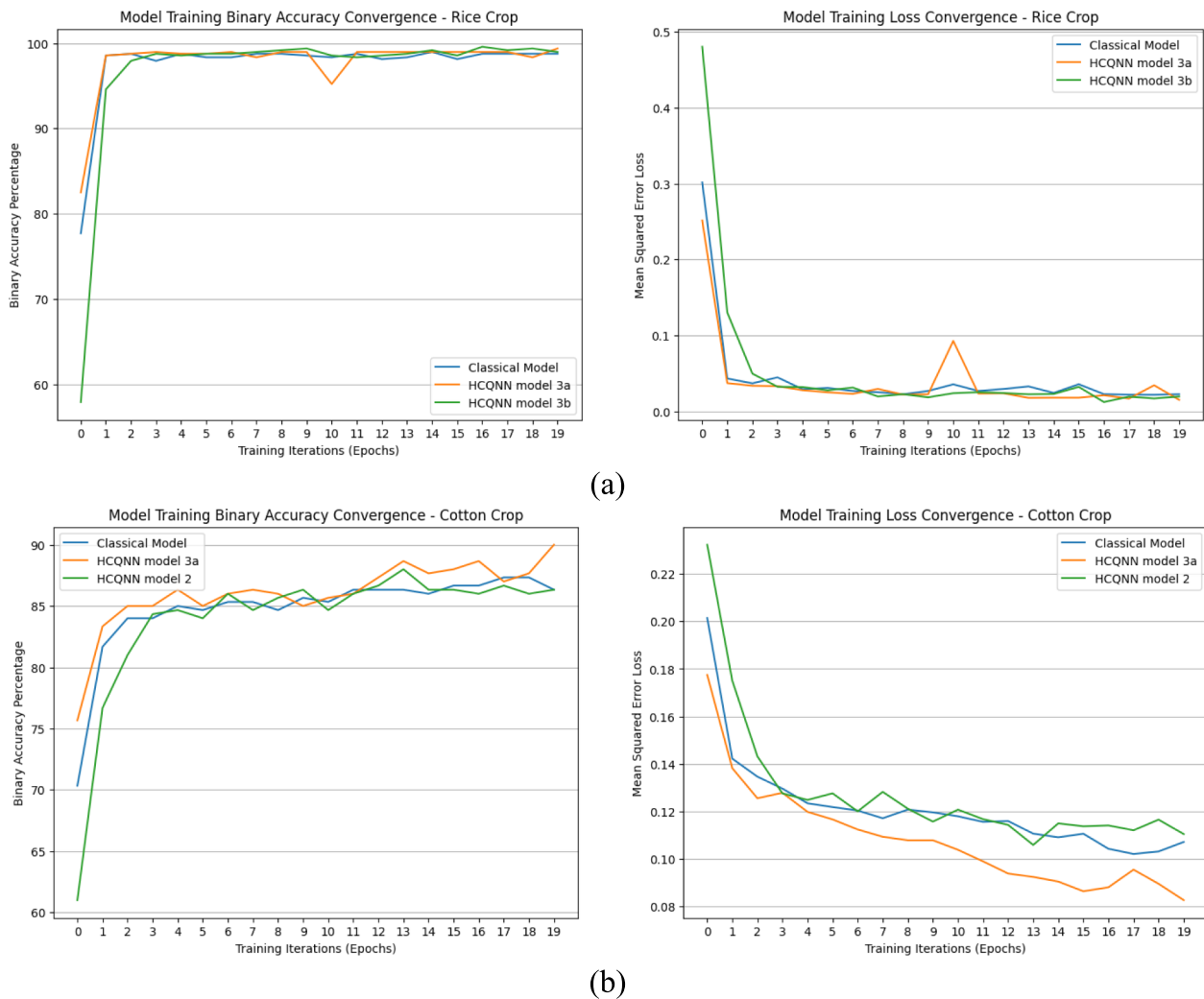
both datasets. The classical MLFFNNs are shown in Fig. 2, and their performance (scores) is shown in Table 3.

The performance of our HCQNNs applied to the same datasets is shown in Table 4, for all four HCQNN subcategories. A detailed presentation of all the results acquired during the HCQNN models’ evaluation is shown in Tables B1–B16 in the Supplementary Information. Furthermore, both the HCQNN and the classical MLFFNN models’ F1 score variance is demonstrated in Fig. 13.

The best performing Feature Map and Ansatz combination in subcategories 1, 3a, and 3b was the “ZZFeatureMap” and “RealAmplitudes” one, whereas in subcategory 2, the “AngleEmbedding” and Circuit 15 was the best performing overall. The results obtained during the HCQNN models’ evaluation indicate that even though HCQNN models with circuits 14 and 15, when utilized either as Feature Mapping circuits or as Ansätze, are able to perform generally quite well, in most cases, their efficacy is not better when compared to the other HCQNN model structures. Thus, it led us to the conclusion that, for the datasets considered within this research, the expressibility of these variational quantum circuits did not prove beneficial for the HCQNN models’ performance.

We also compared the performance of our HCQNN models to the performance of quantum support vector





**Fig. 11** Best HCQNN and classical MLFFNN models' training accuracy and loss curves for 20 training epochs: **a** "Rice Crop" dataset and **b** "Cotton Crop" dataset

machine classifier. We utilized Qiskit's quantum support vector machine classifier (QSVC) to classify the data in both datasets. The performance of the QSVC for the "Rice Crop" and "Cotton Crop" datasets is shown in Tables 5 and 6, respectively. As part of these tests, we experimented with different kernels and studied their impact to the QSVC's performance. We also tested the performance of both the "ZZFeatureMap" and the "AngleEmbedding" as Feature Mapping circuits.

By comparing the performance of Tables 2, 3, 4, 5, and 6, it is evident that the HCQNN models were able to outperform both their classical MLFFNN counterparts and the QSVC, for all the subcategories considered and for both datasets tested against. For the "Rice Crop" dataset, the best performing HCQNN model matches the best score

produced by the classical models, as seen in Table 2. For the "Cotton Crop" dataset, the best performing HCQNN model surpasses the corresponding performance of the classical models used for comparison, except for the "GradientBoostingClassifier," "BaggingClassifier," and the "RandomForestClassifier" classical models. A possible reasoning behind the observed high performance of the HCQNN models, especially when compared with their classical counterparts and other classical models, is the ability of qubits to exploit a bigger search space, i.e., the Hilbert space, to acquire better solutions. Consequently, when quantum circuits substitute classical neurons within a MLFFNN, thus acting as quantum neurons, they are able to provide better outputs than the classical ones, hence improving the model's overall performance.

**Table 2** Performance of well-known classical models applied to both datasets

| ID | Classical models              | “Rice Crop” F1 score | “Cotton Crop” F1 score |
|----|-------------------------------|----------------------|------------------------|
| 1  | GradientBoostingClassifier    | 0.8065               | 0.8925                 |
| 2  | BaggingClassifier             | 0.9852               | 0.8830                 |
| 3  | SVC                           | 1.0000               | 0.75                   |
| 4  | RandomForestClassifier        | 0.8130               | 0.8947                 |
| 5  | KNeighborsClassifier          | 0.9852               | 0.8350                 |
| 6  | AdaBoostClassifier            | 0.9804               | 0.8488                 |
| 7  | LogisticRegression            | 0.9417               | 0.8469                 |
| 8  | GaussianNB                    | 0.8024               | 0.8543                 |
| 9  | DecisionTreeClassifier        | 0.9852               | 0.8280                 |
| 10 | XGBClassifier                 | 0.9569               | 0.8750                 |
| 11 | MLPClassifier                 | 1.0000               | 0.8634                 |
| 12 | GaussianProcessClassifier     | 0.9709               | 0.8241                 |
| 13 | QuadraticDiscriminantAnalysis | 0.8372               | 0.8718                 |
| 14 | SGDClassifier                 | 0.9447               | 0.8400                 |
| 15 | ExtraTreesClassifier          | 0.9662               | 0.8718                 |

## 5 Conclusion

In this paper, we introduced hybrid classical-quantum neural networks in which quantum neurons were integrated in the hidden layers. Our method is general and can be used to introduce one or more quantum neurons into classical neural networks. We showed that the integration of even one quantum neuron enhances the performance of the classical neural network. We developed, trained, and evaluated all possible HCQNN three and four hidden layer architectures using two

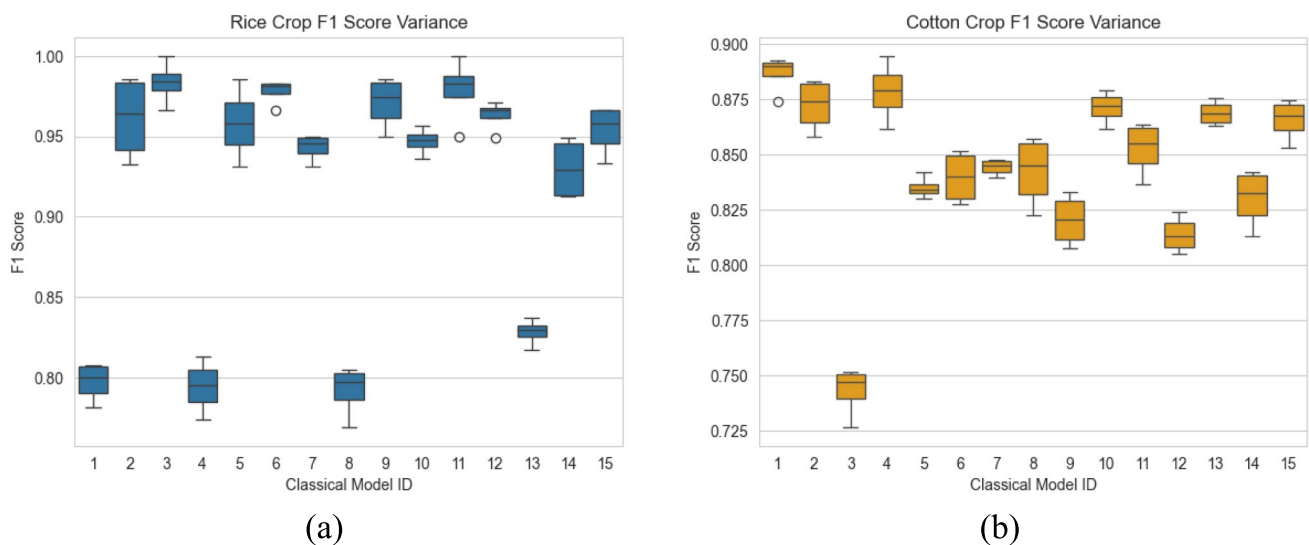
**Table 3** Performance of the two classical MLFFNN architectures

| Classical MLFFNN architecture                                                                  | “Rice Crop” F1 score | “Cotton Crop” F1 score |
|------------------------------------------------------------------------------------------------|----------------------|------------------------|
| <b>Model 1:</b> Input $\rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ | 0.9804               | 0.8615                 |
| <b>Model 2:</b> Input $\rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$                | 0.9756               | 0.8629                 |

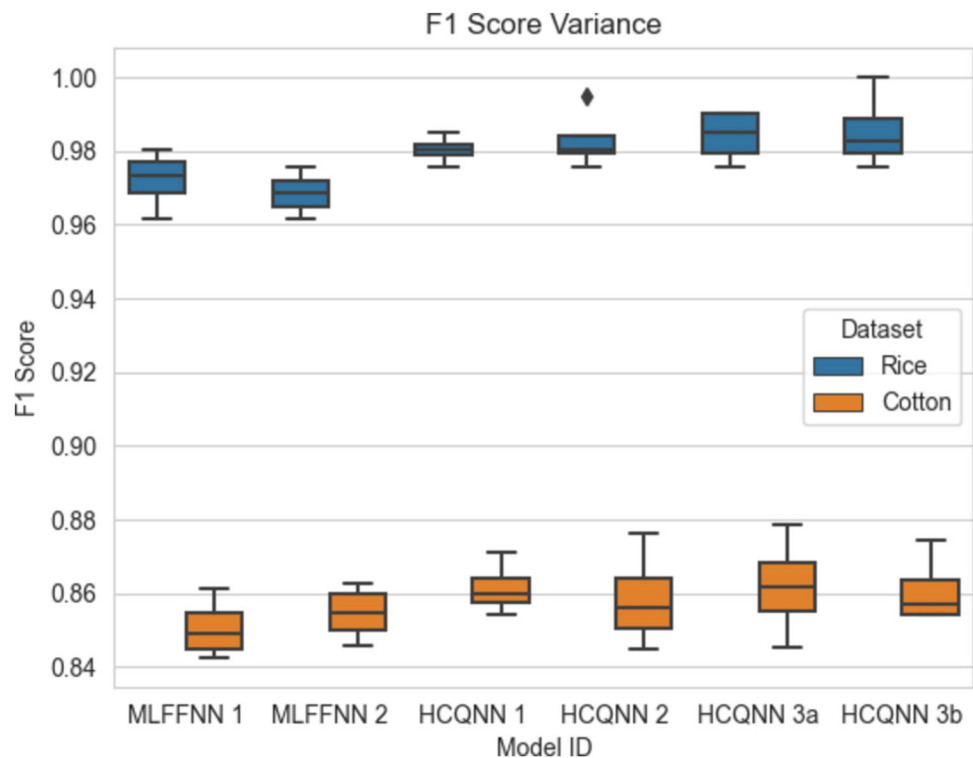
**Table 4** Performance of our HCQNN architecture subcategories applied to the same datasets

| HCQNN subcategory | “Rice Crop” F1 score | “Cotton Crop” F1 score |
|-------------------|----------------------|------------------------|
| 1                 | 0.9852               | 0.8713                 |
| 2                 | 0.995                | 0.8763                 |
| 3a                | 0.9901               | 0.8788                 |
| 3b                | 1.0                  | 0.8744                 |

remote sensing datasets. We compared the performance of the HCQNNs with the performance of well-known classical neural networks for the same datasets. We also compared the performance of the HCQNNs with the performance of the QSVC for the same datasets. We have demonstrated that quantum circuits, when used either as quantum layers or as quantum neurons integrated into a multilayer feedforward neural network, can improve its performance. We also tested several combinations of Feature Mapping circuits and Ansätze, and our results indicate the most consistent and best performing was the “ZZFeatureMap” and “RealAmplitudes” quantum circuit combinations. The number of qubits required for HCQNNs depends on the dataset features or the previous neural network layer size. Our architecture is

**Fig. 12** Classical models' F1 scores variance comparison on **a** “Rice Crop” dataset and **b** “Cotton Crop” dataset

**Fig. 13** Classical MLFFNN and HCQNN models' F1 scores variance comparison on **a** "Rice Crop" dataset and **b** "Cotton Crop" datasets



**Table 5** QSVC performance evaluation for the "Rice Crop" dataset. The best results are shown

| "Rice Crop"—QSVC F1 score                     |                           |                            |
|-----------------------------------------------|---------------------------|----------------------------|
| Kernel                                        | ZZ Feature Map (5 qubits) | Angle Embedding (3 qubits) |
| FidelityQuantumKernel                         | 0.877                     | 0.9394                     |
| TrainableFidelityQuantumKernel                | 0.8729                    | 0.9394                     |
| FidelityQuantumKernel (gamma = 15.0)          | 0.8729                    | 0.9394                     |
| TrainableFidelityQuantumKernel (gamma = 15.0) | 0.8729                    | 0.9394                     |

**Table 6** QSVC performance evaluation for the "Cotton Crop" dataset. The best results are shown

| "Cotton Crop"—QSVC F1 score                   |                           |                            |
|-----------------------------------------------|---------------------------|----------------------------|
| Kernel                                        | ZZ Feature Map (7 qubits) | Angle Embedding (3 qubits) |
| FidelityQuantumKernel                         | 0.75                      | 0.7464                     |
| TrainableFidelityQuantumKernel                | 0.75                      | 0.7464                     |
| FidelityQuantumKernel (gamma = 15.0)          | 0.75                      | 0.7464                     |
| TrainableFidelityQuantumKernel (gamma = 15.0) | 0.75                      | 0.7464                     |

scalable, easy to implement, and enhances the capabilities of MLFFNNs. The HCQNN architecture can be used to classify remote sensing data and as a general-purpose classifier.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s42484-024-00230-8>.

**Acknowledgements** The views reflected in this article are the views of the author(s) and do not necessarily reflect the views of the global EY organization or its member firms.

**Author Contribution** All authors wrote and reviewed the manuscript.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing interests** The authors declare no competing interests.

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